

Supporting Information for “Per-site meter-scale regolith thermal conductivity at the Apollo 15 and 17 boreholes”

Contents of this file: Text S1–S2, Figures S1–S2, Table S1 (to be populated on co-author approval of the moves listed in the header comment of this file).

Introduction

This Supporting Information contains extended methodological detail and diagnostic material referenced from the main text. Figure S1 is the per-site density-scalar diagnostic under the Martínez and Siegler (2021) $K(T, \rho)$ form, cited from §4.3 of the main text.

References

- W. D. Carrier, G. R. Olhoeft, and W. Mendell. Physical properties of the lunar surface. In G. Heiken, D. Vaniman, and B. French, editors, *Lunar Sourcebook: A User’s Guide to the Moon*, pages 475–594. Cambridge University Press, 1991.
- A. Martínez and M. A. Siegler. A global thermal conductivity model for the surface of the Moon. *Journal of Geophysical Research: Planets*, 126(11):e2021JE006829, 2021. doi: 10.1029/2021JE006829.
- J. K. Mitchell, W. N. Houston, W. D. Carrier, and N. C. Costes. Apollo soil mechanics experiment S-200, final report. *NASA Contractor Report (NASA-CR-134306), Space Sciences Laboratory Series*, 15(7), 1973. Bulk density 1.4–1.8 g/cm³; porosity 0.30–0.45 across Apollo cores.
- M. A. Wieczorek, B. L. Jolliff, A. Khan, M. E. Pritchard, B. P. Weiss, J. G. Williams, L. L. Hood, K. Richter, C. R. Neal, C. K. Shearer, I. S. McCallum, S. Tompkins, B. R. Hawke, C. Peterson, J. J. Gillis, and B. Bussey. The constitution and structure of the lunar interior. In *Reviews in Mineralogy and Geochemistry*, volume 60, pages 221–364. 2006. doi: 10.2138/rmg.2006.60.3.

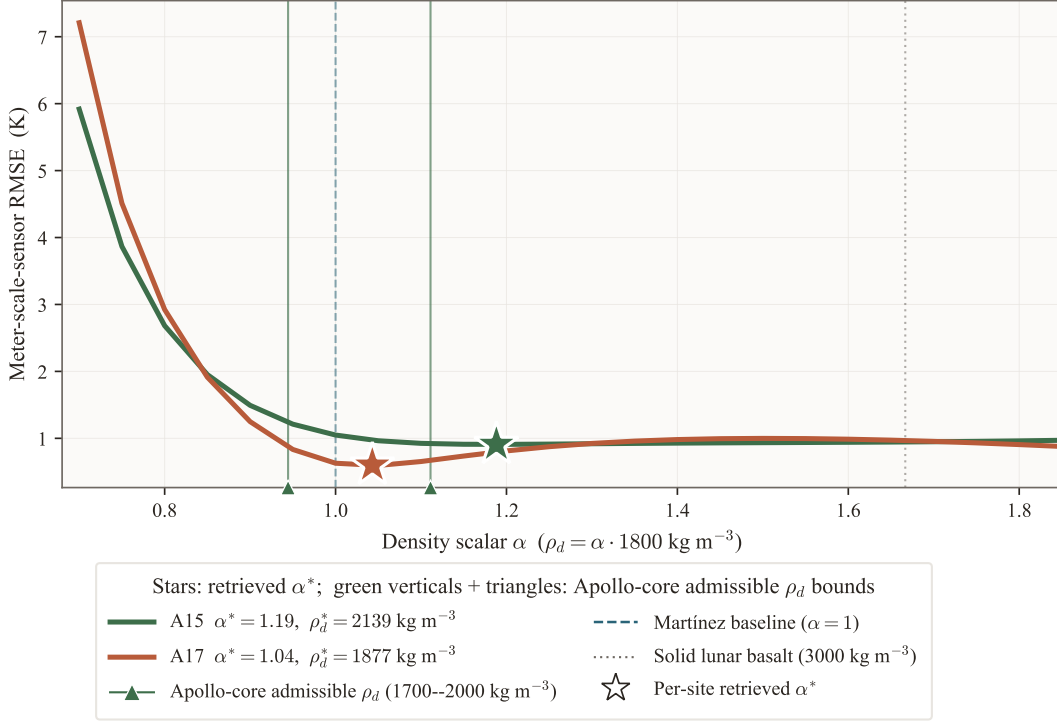


Figure S1: Diagnostic per-site retrieval under the Martínez and Siegler (2021) $K(T, \rho)$ conductivity form. Meter-scale-sensor RMSE as a function of the per-site density scalar α (main-text eq. 9), for both sites, with the retrieved α^* starred. The scalar α rescales the asymptotic (meter-scale) bulk density entering the Martínez model, $\rho_d^{\text{site}} = \alpha \cdot 1800 \text{ kg m}^{-3}$, leaving the four Martínez coefficients fixed. Dashed teal: published Martínez baseline ($\alpha = 1$, $\rho_d = 1800 \text{ kg m}^{-3}$). Green verticals: Apollo-core admissible density range ($\rho_d = 1700\text{--}2000 \text{ kg m}^{-3}$; Mitchell et al., 1973; Carrier et al., 1991). Dotted gray: solid-basalt upper bound (3000 kg m^{-3} ; Wieczorek et al., 2006). Both $\text{RMSE}(\alpha)$ bowls bottom inside or at the edge of the admissible band ($\alpha_{A15}^* = 1.19$, $\rho_d^* \approx 2139 \text{ kg m}^{-3}$; $\alpha_{A17}^* = 1.04$, $\rho_d^* \approx 1877 \text{ kg m}^{-3}$), so the published $K(T, \rho)$ is physically admissible at both boreholes. The flat A17 bowl also shows that no admissible density adjustment closes the residual factor-of- ~ 1.5 RMSE gap to the per-site Hayne fit; sharpening the A17 gradient further requires freedom in the conductivity mechanism (the contact/radiative split), not in the density input.